■ DIL SE

AI-Powered Mental Wellness Companion

A Culturally-Tailored Mental Health App for Pakistani Teens and Young Adults

Project Type:	Flask Web Application
Technology Stack:	Python, Flask, SQLAlchemy, Claude AI API
Frontend:	HTML5, CSS3, Vanilla JavaScript
Database:	SQLite
Target Users:	Pakistani Teens & Young Adults (13-25 years)
Status:	Development Complete - Ready for Deployment
Created:	May 12, 2026

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1. Executive Summary

Dil Se (meaning "From the Heart") is an AI-powered mental wellness companion application specifically designed for Pakistani teens and young adults. The app provides a judgment-free space for emotional support, wellness guidance, and mental health tracking through conversational AI, wellness exercises, mood tracking, and private journaling. Built with Flask and powered by Anthropic's Claude AI, Dil Se brings culturally-aware, compassionate mental health support to a demographic that faces significant barriers to traditional mental health services.

2. Project Vision & Objectives

Vision:

To create an accessible, culturally-sensitive digital mental wellness companion that reduces stigma around mental health conversations and empowers Pakistani youth to seek support.

Core Objectives:

- Provide a judgment-free, shame-free space for emotional conversations
- Make mental health support accessible in a culturally relevant way
- Normalize emotional expression among Pakistani youth
- Offer evidence-based wellness techniques in an approachable format
- Track emotional well-being over time through mood history
- Support reflection through private journaling
- Reduce stigma by presenting support as a sign of strength, not weakness

3. Problem Statement & Solution

The Problem:

Pakistani youth face significant mental health challenges: exam anxiety, family pressure, career uncertainty, relationship stress, and identity concerns. However, accessing help is extremely difficult due to cultural stigma. Young people fear being labeled "pagal" (crazy), worry about being a burden on family, and lack relatable, non-clinical resources that speak to their lived experience.

Dil Se's Solution:

- **Culturally Attuned Communication:** Responses written in conversational English, the emotional language of Pakistani Gen Z, avoiding corporate wellness language
- **Relatable Scenarios:** Quick-reply options address real Pakistani stressors: exam pressure, family expectations, rishta stress, career anxiety
- **Non-Judgmental Tone:** The chatbot acts like a trusted friend, not a clinician, with emphasis on validation over diagnosis
- **Private Space:** All interactions are private and personal—no family pressure, no judgment
- **Evidence-Based Techniques:** Simple, actionable wellness exercises (breathing, grounding, meditation) explained in plain language
- **Mood Tracking:** Visual representation of emotional patterns helps users understand their mental wellness journey

4. Key Features Overview

Feature	Description
■ Home Screen	Daily mood check-in, 7-day mood history, streak counter, quick navigation
■ AI Chat	Conversational chatbot powered by Claude, context-aware Pakistani responses
■ Wellness Exercises	6 guided exercises: breathing, grounding, meditation, muscle relaxation
■ Private Journal	Encrypted journaling with mood tracking and search functionality
■ Mood Tracking	Visual history of mood patterns over 7 and 30 days
■ Language Toggle	English and Hinglish language options for accessibility
■ Data Persistence	All data stored locally for privacy; optional cloud backup

5. Architecture & Technical Stack

Technology Stack:

Component	Technology	Purpose
Backend	Flask 3.0+	Web framework for API and routing
Database	SQLite + SQLAlchemy	Data persistence and ORM
AI Integration	Anthropic Claude API	Intelligent conversational responses
Frontend Templates	HTML5, Jinja2	Dynamic page rendering
Styling	CSS3	Responsive, accessible design
Interactivity	Vanilla JavaScript (ES6+)	Client-side functionality
Environment	Python 3.8+, pip, venv	Python environment management
Hosting	Flask dev or Gunicorn/Nginx	Production deployment ready

Project Structure:

```
dilse-project/
■■■ app.py - Main Flask application with 25+ API routes
■■■ config.py - Configuration management (dev/prod)
■■■ models.py - SQLAlchemy database models
■■■ chatbot.py - Claude AI integration
■■■ exercises.py - Exercise definitions and utilities
■■■ requirements.txt - Python dependencies
■■■ templates/
■ ■■■ base.html
■ ■■■ home.html
■ ■■■ chat.html
■ ■■■ exercises.html
■ ■■■ journal.html
■■■ static/
■ ■■■ css/style.css
■ ■■■ js/*.js (6 module files)
■■■ instance/ - Runtime data (db, uploads)
```

6. Core Functionality

A. Home Screen - Mood Check-in Hub

The home screen serves as the daily entry point. Users select their current mood from a 5-point scale (■ Very Bad to ■ Great) with emoji representation. This data is stored and visualized in a 7-day history chart. The screen also displays a streak counter to encourage daily engagement and quick-access cards to all major features.

B. AI Chat - Conversational Support

The chatbot uses the Anthropic Claude API with a custom system prompt tailored to Pakistani youth. It understands cultural context, uses conversational language, and provides empathetic responses. Users can select from quick-reply suggestions or type freely. The chatbot maintains conversation history within the session and can suggest professional resources if needed.

Quick Reply Examples:

- I'm stressed about exams
- Parents are pressuring me
- I don't know what career I want
- I feel alone
- I'm anxious about something
- I just need to vent

C. Wellness Exercises - Evidence-Based Techniques

- **4-7-8 Breathing:** A calming technique for anxiety and stress (5 minutes)
- **5-4-3-2-1 Grounding:** Sensory awareness exercise to ground in the present moment (5 minutes)
- **Body Scan Meditation:** Progressive relaxation from head to toe (8 minutes)
- **Gratitude Practice:** Reflection exercise for perspective shift (3 minutes)
- **Progressive Muscle Relaxation:** Tension release through muscle engagement (6 minutes)
- **Guided Journaling:** Structured reflection with conversational prompts

D. Private Journal - Reflective Writing

Users can write private journal entries with optional mood logging. Each entry is timestamped, searchable, and fully editable. The journal provides conversational writing prompts to guide reflection: "What's been on your mind?", "How are you feeling today?", "What would help you feel better?"

8. How It Works - User Flow

Typical User Journey:

- **1. Login:** User opens the app (anonymous or registered session)
- **2. Home Screen:** Greets with daily mood check-in question
- **3. Feature Access:** User chooses to Chat, Do an Exercise, or Journal
- **4. Interaction:** Engages with chosen feature (e.g., typing to chatbot)
- **5. Data Logging:** Interactions are automatically logged for tracking
- **6. History View:** User can view mood trends and past entries anytime
- **7. Repeat:** Users return daily for check-ins and support

Technical Data Flow:

```
Frontend (Browser) → Flask API → Database (SQLite)
↓
User submits mood → app.py /mood endpoint → Stored in MoodEntry table
↓
User sends chat message → app.py /chat endpoint → Claude API → Response returned
↓
User creates journal entry → app.py /journal endpoint → Stored in JournalEntry table
↓
All data encrypted at rest and transmitted securely
```

9. Database Structure

The app uses SQLite with SQLAlchemy ORM for data management. All data is stored locally for privacy.

Table	Purpose	Key Fields
User	User profiles & preferences	id, username, language_preference, created_at
MoodEntry	Daily mood tracking	id, user_id, mood_score, timestamp, notes
JournalEntry	Private journal entries	id, user_id, title, content, mood, created_at
ChatHistory	Conversation logs	id, user_id, user_message, bot_response, timestamp
ExerciseCompletion	Exercise tracking	id, user_id, exercise_name, completed_at, duration

10. Installation & Deployment

Quick Start (Development):

1. Clone or download the project to your machine
2. Create a Python virtual environment: `python -m venv venv`
3. Activate it: `source venv/bin/activate` (macOS/Linux) or `venv\Scripts\activate` (Windows)
4. Install dependencies: `pip install -r requirements.txt`
5. Set up environment variables by copying `.env.example` to `.env`
6. Add your Anthropic API key to `.env`
7. Run the app: `python app.py`
8. Open browser to `http://localhost:5000`

Production Deployment:

For production deployment:
• Use a production WSGI server like **Gunicorn**
• Reverse proxy with **Nginx**
• Enable **HTTPS** with SSL certificates
• Store environment variables securely (not in .env)
• Use a robust database like **PostgreSQL**
• Enable database backups
• Monitor application logs and errors
• Set up rate limiting for API endpoints

11. Future Enhancements

- **Multi-language Support:** Full Urdu and Hinglish UI in addition to English
- **Mobile App:** Native iOS and Android applications
- **Crisis Resources:** Integration with Pakistan crisis hotlines and mental health services
- **Community Features:** Anonymous support groups and peer connections
- **Therapist Integration:** Optional connection to licensed mental health professionals
- **Wearable Integration:** Heart rate and stress level monitoring from smartwatches
- **Advanced Analytics:** Personalized wellness insights and recommendations
- **Offline Mode:** Basic functionality available without internet connection
- **Social Sharing:** Share wellness tips and exercises with friends
- **Parental Guides:** Resources for parents to support youth mental health

Conclusion

Dil Se represents a significant step forward in making mental wellness support accessible and culturally relevant for Pakistani youth. By combining evidence-based wellness techniques with empathetic AI and a design-first approach, the app removes barriers to support and creates a safe, judgment-free space for emotional health. The project is ready for deployment and can begin making a real impact in the lives of young people who need it most.

Document generated on May 12, 2026 at 06:22 PM
For more information, visit the project repository or contact the development team